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Phenological matching rather than genetic variation in host preference underlies geographical variation in host plants used by orange tip butterflies

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An insect species that shows variation in host species association across its geographical range may do so either because of local adaptation in host plant preference of the insect or through environmentally or genetically induced differences in the plants, causing variation in host plant suitability between regions. In the present study, we experimentally investigate the host plant preference of *Anthocharis cardamines* (orange tip butterfly) in two populations from the UK and two from Sweden. Previous reports indicate that *A. cardamines* larvae are found on different host plant species in different regions of the UK, and some variation has been reported in Sweden. Host plant choice trials showed that females prefer to oviposit on plants in an earlier phenological stage, as well as on larger plants. When controlling for plant phenological stage and size, the host species had no statistically significant effect on the choice of the females. Moreover, there were no differences in host plant species preference among the four butterfly populations. Based on our experiment, the oviposition choice by *A. cardamines* mainly depends on the phenological stage and the size of the host plant. This finding supports the idea that the geographical patterns of host–plant association of *A. cardamines* in the UK and Sweden are consequences of the phenology and availability of the local hosts, rather than regional genetic differences in the host species preference of the butterfly. © 2016 The Linnean Society of London, *Biological Journal of the Linnean Society*, 2016, **119**, 1060–1067.

KEYWORDS: *Alliaria petiolata* – *Anthocharis cardamines* – *Cardamine pratensis* – host plant preference – oviposition – plant–herbivore interaction.

INTRODUCTION

The interaction between insects and their host plants has long fascinated ecologists (Ehrlich & Raven, 1964; Gilbert & Raven, 1975; Scriber & Feeny, 1979). The host–plant associations of insect species depend on life-history traits, the phylogenetic history of host association, and plant traits, as well as a large number of environmental factors (Ehrlich & Raven, 1964; Gilbert & Raven, 1975; Feeny, 1976; Futuyma & Moreno, 1988; Singer & McBride, 2012). Insects can vary in their host use across time as well as space, and exhibit a geographical mosaic of association with different host plant species (Zovi *et al.*, 2008; Wiklund & Friberg, 2009; Singer & McBride, 2012). Local adaptation to host plant species and populations appears to be relatively common in herbivorous insects (Mopper, 1996; Kuussaari, Singer & Hanski, 2000; Gotthard, Margraf & Rahier, 2004; Zovi *et al.*, 2008; Garrido, Andraca-Gómez & Fornoni, 2012). Such local adaptation should be detectable in common garden or transplantation experiments as differences in insect preference and performance for different host plants (Garrido *et al.*, 2012) and may give rise to geographical variation in host species use (Feder, Hunt & Bush, 1993; Singer & Parmesan, 1993).

However, geographical variation in host use need not necessarily come about through genetic

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differences in host plant preference (Gotthard *et al.*, 2004; Ferrari *et al.*, 2006) but, instead, can be caused by environmental variation across a geographical range, resulting in plant species varying in suitability; for example, through variation in phenological stage, abundance or size across an insect's range (Kuussaari *et al.*, 2000). Additionally, genetic variation in a plant species across an insect's range may cause variation in host species use (Singer & Parmesan, 1993). Local variation in plant abundance, size, and phenology may also cause variation in host plant use (Thomas, 1987; Kerpel & Moreira, 2005). In the present study, we use oviposition choice trials with butterflies from different geographical origins to examine what controls butterfly host plant use across its range in two countries (Courtney & Duggan, 1983; Navarro-Cano *et al.*, 2015).

By selecting which plant to oviposit on, a female butterfly determines the environment that her offspring will encounter. A multitude of variables, such as plant species, size, microhabitat, and phenological (equivalent to developmental in the present study) stage of the plant, can impact the fitness of her offspring by affecting the quality and quantity of food available, as well as the parasitism rate and competition (Wiklund & Åhrberg, 1978; Mayhew, 1997). In addition to the factors relating to the plant, a female's own state may influence her host plant choice. For example, egg load, search time, and experienced abundance and quality of different host plant species can influence the oviposition choice (Minkenberg, Tatar & Rosenheim, 1992; West & Cunningham, 2002; Singer, 2015).

Anthocharis cardamines (orange tip butterfly) has been described as both oligophagous and polyphagous (Wiklund & Åhrberg, 1978; Wiklund & Friberg, 2009; Navarro-Cano *et al.*, 2015) (Fig. 1). The larvae feed mainly on Brassicacae and utilize most of the available species of Brassicacae in the UK and Sweden (Wiklund & Åhrberg, 1978; Courtney, 1982a; Courtney & Duggan, 1983; Wiklund & Friberg, 2009). *Anthocharis cardamines* is considered to be a phenological specialist because the larvae mainly feed on the developing seed pods of its host plants and hence need to complete their development in a time window of the host plants' life cycle (Wiklund & Åhrberg, 1978; Courtney, 1981). Female butterflies use visual cues to find suitable host plants and inspect them more closely at landing (Wiklund & Åhrberg, 1978; Courtney & Duggan, 1983). Their use of a wide range of host plant species within one plant family might be a result of spatial variation, as well as pronounced among-year variation, in the phenological stage of host plants during the flight period of the butterfly (Wiklund & Friberg, 2009). Oviposition in *A. cardamines* has been reported to be time constrained rather than egg constrained (Courtney, 1982a). Female orange tips often reportedly fail to lay all their eggs in their life span, mainly as an effect of forced inactivity during poor weather (Courtney & Duggan, 1983). Such a phenomenon may contribute to the butterfly using multiple host species and being less choosy, despite that growth rate and size of *A. cardamines* varies with host species (Davies & Saccheri, 2013; Posledovich *et al.*, 2015). Previous studies report that *A. cardamines* appears to be associated with *Alliaria petiolata* (garlic mustard) in the south of the UK and *Cardamine pratensis* (cuckooflower) in the north (Courtney & Duggan, 1983). In Sweden, *A. cardamines* has been shown to be associated with different host plants in different regions, although *C. pratensis* stands out as being a major host plant in all regions (Navarro-Cano *et al.*, 2015). In the present study, we investigate whether there is local adaptation in host plant preference or whether the major driver of host plant association in Sweden and the UK is circumstantial, in the sense of resulting from local variation in host plant phenological stage and abundance during the flight season of *A. cardamines*. We do this by

Figure 1. *Anthocharis cardamines* resting on its host plants *Alliaria petiolata* (left) and *Cardamine pratensis* (right). Photographer: Sandra Stålhandske.

examining how host plant choice of females from four localities (two from Sweden and two from the UK) depends on: (1) the population origin of females; (2) host plant species; (3) host plant size; and (4) host plant phenological stage.

MATERIAL AND METHODS

Host plant choice experiments

We investigated oviposition host plant choice in 48 mated female *A. cardamines*. The females were collected as eggs in the wild in 2012 and reared to pupation under common garden conditions in the laboratory with *ad libitum* access to *A. petiolata*. The diapausing pupae were overwintered in the laboratory (at 1–2 °C) and, after adult emergence in spring 2013, they were mated prior to being used in the oviposition experiment. The females were from two latitudinally separate localities in the UK (Bournemouth: 50.72N, 1.88W, $N = 8$; Durham: 54.68N, 1.57W, $N = 11$), where Bournemouth butterflies appear be associated with *A. petiolata* and Durham butterflies with *C. pratensis* in the wild (Courtney & Duggan, 1983). We also had butterflies from two latitudinally separate localities in Sweden (Skåne: 55.49N, 14.05E, $N = 11$; Ångermanland: 63.03N, 18.19E, $N = 12$). In these populations, the host species association is less clear-cut (Wiklund & Friberg, 2009; Navarro-Cano *et al.*, 2015). *Alliaria petiolata* individuals were dug up from a locality at Frescati, Stockholm (59.4N, 18.1E, $N = 68$), in April of 2013, and kept at Tovetorp, Södermanland (58.95N, 17.15E), close to the experimental site. *Cardamine pratensis* ($N = 37$) was collected at the same time at Tullgarn castle (58.95N, 17.60E) and reared at the same site as *A. petiolata*. New individuals of *C. pratensis* were collected during the experiment in close proximity to the experimental site ($N = 21$).

The oviposition choice trials were performed at Kronängen, Södermanland, Sweden (58.97N, 17.16E) in the late spring of 2013. Of the 48 females, 42 laid eggs during their trials. A trial consisted of one mated female being released into an outdoor net cage (6 × 10 × 3 m) with five plants of *A. petiolata* (garlic mustard) and five plants of *C. pratensis* (cuckooflower). When it was estimated that the female had laid five eggs, she was removed from the cage. For each plant in the experimental cage during the oviposition trial, the species, height (cm), and total number of flowers, buds, and seedpods were noted. The phenological stage was estimated as the mean developmental state of plant reproductive structures, where buds were given a stage value of zero, flowers a value of 0.5, and seedpods a value of 1. The phenological stage of a plant could thus range from 0 (all buds) to 1 (all seedpods). Four trials were run in parallel in four cages. The positions of the plants were predetermined and similar in the four cages, although the placement of the individual plants was randomized for each trial. The size and phenological stage of the plants were aimed to cover the entire possible range for each experimental trial but, in practice, we were limited by what was available (Fig. 2, see Supporting Information, Fig. S1). Each plant was used for multiple trials. After each trial, plants were checked for eggs. When found, each egg was removed using a wet paintbrush. This was carried out with the aim of removing chemical signals that could act as a deterrent signal for ovipositing in later trials (Dempster, 1992).

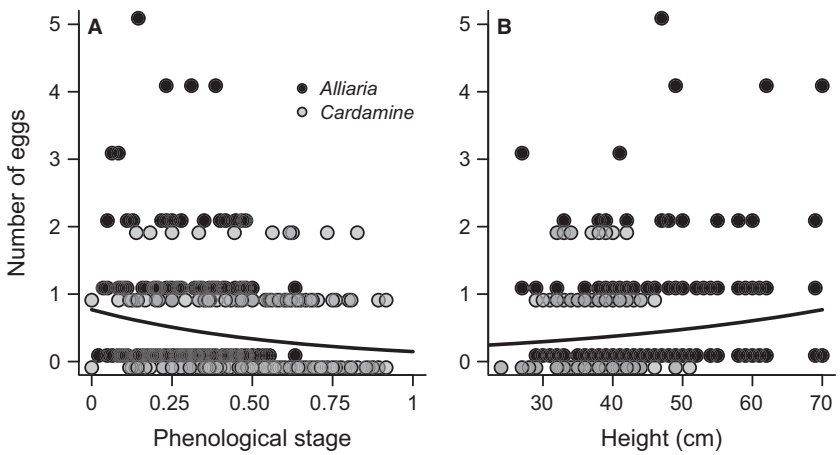

Figure 2. Observed number of eggs laid on the plants. A, observed number of eggs laid on plants of different phenological stages. B, observed number of eggs laid on plants with varying heights. The black and grey points, shifted slightly up and down for clarity, indicate host plant species. The lines show expectations from a Poisson regression.

Statistical analysis

All statistical tests were performed using R, version 3.2.2 (R Core Team, 2015). The number of eggs laid on each plant was used as a response variable and the full model included host plant species, phenological stage, height, population origin of the female, and ambient temperature as fixed effects, as well as female, cage, and individual plant as random effects. We fitted generalized linear models using the lme4 package (Bates *et al.*, 2015) with Poisson errors. Nonsignificant interactions and nonsignificant main effects that were not of interest to our original research questions were removed. The outcome was the same when using either of two different measurements of plant size in the model (total number of buds, flowers, and seedpods, or height). We used plant height in the model because the effect size of this variable was slightly greater than that for the total number of buds, flowers, and seedpods. A quadratic term of the indicator of plant phenological stage was tested but discarded because it did not improve the model fit. To investigate whether females within each locality had different preferences for host plant species, the final model was run with female and the interaction of female and plant species included. To examine whether our method of removing the oviposition deterrent influenced oviposition in later trials, we ran the final model including an explanatory variable indicating whether the plant had been oviposited on in a previous trial or not. Finally, a model with only host plant species as an explanatory variable was fitted for comparison with the final model that included plant phenological stage and size.

For comparison of variance components, the final model was rerun using MCMCglmm (Hadfield, 2010) because this approach estimated a residual variance that corresponds to overdispersion. One way of evaluating the importance of the variance components is to compare them with each other by calculating how much of the total variance they explain that is not explained by the fixed effects (i.e. by dividing the variance of each random effect with the sum of variances of all random effects, including the residual variance). The MCMCglmm was run with a prior with the variance of each variance component set to 1 and nu set to 0.02. The model was run with 2 000 000 iterations, a thinning interval of 1000, and a burn-in of 100 000 to achieve good sampling.

RESULTS

Phenological stage was highly statistically significant with respect to determining host plant choice of *A. cardamines* (Figs 2, 3; Table 1). The model predicted an average of 0.63 additional eggs on a plant with only buds compared to a plant with only seedpods (when holding the height of the plant at the average height used in the experiment) and an average of 0.60 additional eggs for the earliest compared to the latest of the phenological stages used in the experiment (Fig. 3). The size of a plant also had a statistically significantly effect on oviposition choice (Figs 2, 3; Table 1). The model predicted 0.51 additional eggs for the tallest compared to the shortest plant used in the experiment (70 cm vs. 24 cm in height) for a plant of the average phenological stage used in the experiment (Fig. 3). Host plant species and butterfly population, or their interaction, had no statistically significant effect on the oviposition choice of the females (Table 1; see also Supporting Information Fig. S2).

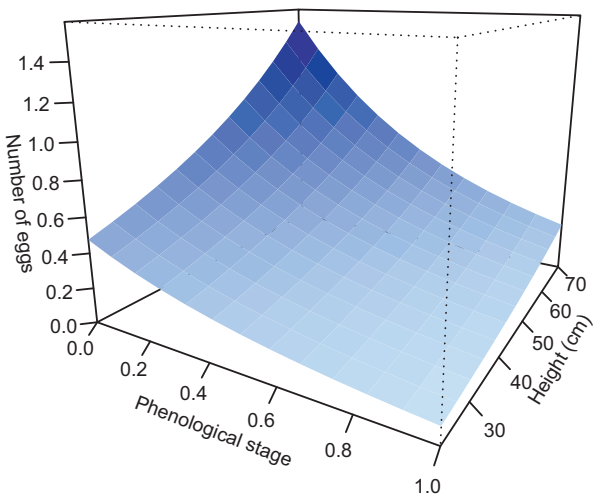

Figure 3. Model predicted number of eggs on plants of different phenological stages and heights.

Table 1. Analysis of oviposition choice in *Anthocharis cardamines* on host plants *Alliaria petiolata* and *Cardamine pratensis*

	χ^2	d.f.	*P*
Plant species	1.81	1	0.179
Population	0.26	3	0.968
Phenological stage	13.31	1	**< 0.001**
Height	6.17	1	**0.013**
Population × Plant species	3.17	3	0.366

Linear model summary of the fixed effects in the final Poisson model of the number of eggs laid. Female ID, cage and plant ID were treated as random effects. Statistically significant *P*-values are shown in bold.

A statistical model with only host plant as explanatory variable showed a significant effect of host plant species ($\chi^2_1 = 6.49$, $P = 0.011$). This effect might have been a result of variation in the phenological stages and heights of the two host plant species used in the experiment (where the *C. pratensis* used had a wider range of phenological stages and generally also later stages), as well as *C. pratensis* plants generally being smaller compared to *A. petiolata* (Fig. 2; see also Supporting Information, Fig. S1). The final analysis, presented above, however, showed that the difference between the two species was a result of differences in phenology and plant size. Oviposition on a plant in a previous trial had no effect on subsequent preference ($\chi^2_1 = 0.05$, $P = 0.826$), suggesting that our method of removing the oviposition deterrent between trials was successful.

Female ID explained most (56.9%) of the variance that was not explained by the fixed effects. Cage explained 15.9%, which is approximately the same proportion as the variance of the residuals (18.2%). Plant ID explained very little of the remaining variance (9.0%) but was judged sufficiently large to be kept in the model. To investigate whether the variation explained by female ID derived from females differing in preference of host species (which would have cofounded our conclusion about location-level preference), female ID was included in the final model as a fixed effect and allowed to interact with host plant species. The outcome was not significant ($\chi^2_{38} = 7.13$, $P = 1$). We thus found no effect of different females having different preferences within or between the different populations.

DISCUSSION

Herbivore–plant interaction

We found no evidence for population differentiation in host plant species preference in *A. cardamines*, although there were strong effects of plant traits on host plant choice. The geographical variation in host plant association found in *A. cardamines* in the UK and Sweden (Courtney & Duggan, 1983; Navarro-Cano *et al.*, 2015) is therefore most likely a result of variation in plant availability and in the matching of butterfly and host phenologies, rather than inherent differences in preference among butterfly populations.

We found that the phenological stage of a host plant is the most important factor for the oviposition choice of female *A. cardamines*. Host plant size also had an effect on oviposition choice. By comparing the final model with the model including only host plant species, what appears as a difference in host species preference can be seen as a consequence of the phenological stage and size of plants. These results support previous findings from field observations regarding host plant choice in *A. cardamines*, where plant developmental stage and size were the main factors affecting oviposition choice (Courtney, 1982a). Studies of host plant use over several years (Courtney, 1982b; Wiklund & Friberg, 2009) have found variation in host species use over time, suggesting that the association is a temporal matching rather than species tracking, fitting well with the findings of the present study. In support of this, plant phenology also strongly affects larval development, survival, and final pupal mass (Courtney, 1981; Davies & Saccheri, 2013; Posledovich *et al.*, 2015). Hence, plant phenological stage likely over-rides the identity of plant species for female host plant choice in *A. cardamines* (Wiklund & Åhrberg, 1978; Courtney & Duggan, 1983; Posledovich *et al.*, 2015). A previous study reported that phenology was more important than host plant species in a feeding choice experiment of a beetle feeding on Passiflora vines (Thomas, 1987). In the present study, the size of a plant also affected oviposition choice, which is in agreement with previous findings (Courtney, 1982a). Survival of larvae has been shown to be greatly reduced if larvae are forced to find another host plant when the one that they hatched on proves insufficient for finishing growth until the fifth instar, which results from a combination of the phenology and size of a plant (Wiklund & Friberg, 2009). This has also been shown in experimentally induced phenological mismatches (Posledovich *et al.*, 2015).

With predicted future climate change (IPCC, 2014), species interactions are vulnerable to change (Visser & Both, 2005). It has been suggested that, rather than measuring phenological change in terms of absolute days, change should be considered in terms of biological interactions (Visser & Both, 2005). Because *A. cardamines* is a phenological specialist and the butterfly relies on matching host plants locally, there may be a change in host plant use at the local level if plants and butterflies react differently to winter and spring warming.

Anthocharis cardamines females use white or otherwise brightly coloured inflorescences as a search cue (Courtney, 1982a). The females in our experiment were attracted to white Apiacae plants directly outside the cages, where the grass had not been cut, as well as to the white flowers of the Brassicacae host plants inside the cage. Once quickly inspected, however, the females would abandon a nonhost plant and continue searching. This agrees well with what has previously been described about the host plant search in *A. cardamines* (Wiklund & Åhrberg, 1978; Courtney, 1982a; Wiklund, 1984). Once a female in the present study had landed on a host plant for the first

time, she would resume flight and spend considerable time inspecting potential host plants before rejecting or accepting a plant for oviposition. Although the white or pale pink flowers act as a visual cue attracting the female, in our experiment, females could encounter a *C. pratensis* with only buds. On one instance, a female landed on such a plant and directly oviposited on it. In a natural setting, it is unlikely that females would oviposit on plants with only buds because they are unlikely to find them. Our linear negative relationship between phenological stage and oviposition preference may therefore differ from what is found in the field, where ovipositing on plants early in their development is unlikely as a result of a lack of detection.

Egg load is known to affect oviposition choice (Minkenberg *et al.*, 1992; West & Cunningham, 2002). Because of weather variation during the experiment, some females had mated for a longer time before being let into the experiment cages. These females may have more readily accepted host plants. However, any such effect in our experiments should have been minimized by the removal of females from the cage once they were estimated to have laid five eggs. There should still have been an opportunity for choice among the ten host plants available in the cage. The temperature during the trial was logged but was found to have no statistically significant importance in our analyses. Other studies have suggested that oviposition choice trials should be carried out sequentially for the acceptance/rejection of individual plants or species (Singer, 1986). These methods may be appropriate for clutch layers because they are very reliant on plant condition. For *A. cardamines*, which lays eggs singly and on different plants, we consider that the experimental set-up we used is appropriate. Furthermore, the use of multiple plants of each host species minimizes the effects of individual plants on butterfly preference.

It has been suggested that the observed associations between *A. cardamines* and its host plant across the UK may to some degree result from a size effect of feeding on different host plants (adult *A. cardamines* that fed on *C. pratensis* as larvae are smaller than *A. cardamines* individuals that fed on *A. petiolata*), which could influence dispersal ability and developmental time. The suggestion is that these effects, when coupled with how the plant species are dispersed in time and space, add to the geographical variation in host plant use (Davies & Saccheri, 2013). The life-history and fitness effects caused by host plant species may be numerous and are unaccounted for in the present study. Previous studies suggest varying the performance of *A. cardamines* on different host plants (Wiklund & Åhrberg, 1978; Davies & Saccheri, 2013), with *C. pratensis* being a superior host in terms of the survival of larvae and low rates of parasitism (Wiklund & Åhrberg, 1978). Fecundity, however, is higher for *A. cardamines* females that grew up on *A. petiolata* compared to *C. pratensis* (Davies & Saccheri, 2013). Additional data on the performance and life-history traits of the four populations on the two investigated host species could have enabled a useful connection between the preference test and potential fitness consequences. However, Posledovich *et al.* (2015) reported that performance was affected highly by host phenological stage, providing a link between the preference of *A. cardamines* females and the performance of their offspring.

In summary, we found that the biggest factor in oviposition choice in *A. cardamines* is the phenological stage of the host plant. The size of a plant also matters, although the effect is smaller. Based on our experiment, the geographical association of host plant species and butterfly in the field across the UK and Sweden is more likely to be a result of phenological matching and variation in host plant abundance than of local adaptation in host plant preferences of the butterfly populations.

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AUTHOR CONTRIBUTIONS

SS, MO, KG, JE, CW, and OL conceived and designed the experiment. SS, MO, and CW performed the experiment. SS and OL analyzed the data. SS wrote the manuscript. KG, JE, CW, and OL reviewed and commented on the manuscript.

REFERENCES

Bates D, Mächler M, Bolker B, Walker S. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* **67:** 1–48.

Courtney S. 1981. Coevolution of pierid butterflies and their cruciferous foodplants. 3. *Anthocharis cardamines* (L) survival, development and oviposition on different hostplants. *Oecologia* **51:** 91–96.

Courtney SP. 1982a. Coevolution of pierid butterflies and their cruciferous foodplants IV. Crucifer apparency and

Anthocharis cardamines (L.) oviposition. *Oecologia* **52:** 258–265.

Courtney SP. 1982b. Coevolution of pierid butterflies and their cruciferous foodplants V. Habitat selection, community structure and speciation. *Oecologia* **54:** 101–107.

Courtney SP, Duggan AE. 1983. The population biology of the orange tip butterfly *Anthocharis cardamines* in Britain. *Ecological Entomology* **8:** 271–281.

Davies WJ, Saccheri IJ. 2013. Maintenance of body-size variation and host range in the orange-tip butterfly: evidence for a trade-off between adult life-history traits. *Ecological Entomology* **38:** 49–60.

Dempster JP. 1992. Evidence of an oviposition-deterring pheromone in the orange-tip butterfly, *Anthocharis cardamines* (L). *Ecological Entomology* **17:** 83–85.

Ehrlich PR, Raven PH. 1964. Butterflies and plants: a study in coevolution. *Evolution* **18:** 586.

Feder JL, Hunt TA, Bush L. 1993. The effects of climate, host plant phenology and host fidelity on the genetics of apple and hawthorn infesting races of *Rhagoletis pomonella*. *Entomologia Experimentalis et Applicata* **69:** 117–135.

Feeny P. 1976. Plant apparency and chemical defense. In: Wallace JW, Mansell RL, eds. *Biochemical interaction between plants and insects*. Berlin: Springer, 1–40.

Ferrari J, Godfray HCJ, Faulconbridge AS, Prior K, Via S. 2006. Population differentiation and genetic variation in host choice among pea aphids from eight host plant genera. *Evolution* **60:** 1574–1584.

Futuyma DJ, Moreno G. 1988. The evolution of ecological specialization. *Annual Review of Ecology and Systematics* **19:** 207–233.

Garrido E, Andraca-Gómez G, Fornoni J. 2012. Local adaptation: simultaneously considering herbivores and their host plants. *New Phytologist* **193:** 445–453.

Gilbert LE, Raven PH. 1975. *Coevolution of animals and plants: Symposium V, First International Congress of Systematic and Evolutionary Biology, Boulder, Colorado, August 1973*: [papers]. Austin, TX: University of Texas Press.

Gotthard K, Margraf N, Rahier M. 2004. Geographic variation in oviposition choice of a leaf beetle: the relationship between host plant ranking, specificity, and motivation. *Entomologia Experimentalis et Applicata* **110:** 217–224.

Hadfield JD. 2010. MCMC methods for multi-response generalized linear mixed models: the MCMCglmm R package. *Journal of Statistical Software* **33:** 1–22.

IPCC. 2014. *Climate change 2013: the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change*. Cambridge: Cambridge University Press.

Kerpel SM, Moreira GRP. 2005. Absence of learning and local specialization on host plant selection by *Heliconius erato*. *Journal of Insect Behavior* **18:** 433–452.

Kuussaari M, Singer M, Hanski I. 2000. Local specialization and landscape-level influence on host use in an herbivorous insect. *Ecology* **81:** 2177–2187.

Mayhew PJ. 1997. Adaptive patterns of host-plant selection by phytophagous insects. *Oikos* **79:** 417.

Minkenberg OPJM, Tatar M, Rosenheim JA. 1992. Egg load as a major source of variability in insect foraging and oviposition behavior. *Oikos* **65:** 134–142.

Mopper S. 1996. Adaptive genetic structure in phytophagous insect populations. *Trends in Ecology & Evolution* **11:** 235–238.

Navarro-Cano JA, Karlsson B, Posledovich D, Toftegaard T, Wiklund C, Ehrlén J, Gotthard K. 2015. Climate change, phenology, and butterfly host plant utilization. *Ambio* **44:** 78–88.

Posledovich D, Toftegaard T, Wiklund C, Ehrlén J, Gotthard K. 2015. The developmental race between maturing host plants and their butterfly herbivore – the influence of phenological matching and temperature. *Journal of Animal Ecology* **84:** 1690–1699.

R Core Team. 2015. *R: a language and environment for statistical computing*. Vienna: R Foundation for Statistical Computing.

Scriber JM, Feeny P. 1979. Growth of herbivorous caterpillars in relation to feeding specialization and to the growth form of their food plants. *Ecology* **60:** 829–850.

Singer MC. 1986. The definition and measurement of oviposition preference in plant-feeding insects. In: Miller JR, Miller TA, eds. *Insect–plant interactions*. New York, NY: Springer, 65–94.

Singer MC. 2015. Adaptive and maladaptive consequences of 'matching habitat choice:' lessons from a rapidly-evolving butterfly metapopulation. *Evolutionary Ecology* **29:** 905–925.

Singer MC, McBride CS. 2012. Geographic mosaics of species' association: a definition and an example driven by plant–insect phenological synchrony. *Ecology* **93:** 2658–2673.

Singer MC, Parmesan C. 1993. Sources of variations in patterns of plant–insect association. *Nature* **361:** 251–253.

Thomas CD. 1987. Behavioural determination of diet breadth in insect herbivores: the effect of leaf age on choice of host species by beetles feeding on Passiflora vines. *Oikos* **48:** 211–216.

Visser ME, Both C. 2005. Shifts in phenology due to global climate change: the need for a yardstick. *Proceedings of the Royal Society of London Series B, Biological Sciences* **272:** 2561–2569.

West SA, Cunningham JP. 2002. A general model for host plant selection in phytophagous insects. *Journal of Theoretical Biology* **214:** 499–513.

Wiklund C. 1984. Egg-laying patterns in butterflies in relation to their phenology and the visual apparency and abundance of their host plants. *Oecologia* **63:** 23–29.

Wiklund C, Åhrberg C. 1978. Host plants, nectar source plants, and habitat selection of males and females of *Anthocharis cardamines* (Lepidoptera). *Oikos* **31:** 169–183.

Wiklund C, Friberg M. 2009. The evolutionary ecology of generalization: among-year variation in host plant use and offspring survival in a butterfly. *Ecology* **90:** 3406–3417.

Zovi D, Stastny M, Battisti A, Larsson S. 2008. Ecological costs on local adaptation of an insect herbivore imposed by host plants and enemies. *Ecology* **89:** 1388–1398.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article:

Figure S1. Illustration of the range of host plant variation in phenological stage, height, and another size indicator (total number of buds, flowers and seed pods) for the two species across the experiment.
Figure S2. Proportion of eggs laid on the two species of host plant used in the experiment from each of the four butterfly populations by day of experiment.